A

PROJECT REPORT

On

Design and Implementation of a Hostel Mess Management System using MySQL

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1. Introduction

Manipal University Jaipur operates two hostel messes, namely BluDove and BluSpring. Under the existing arrangement, every resident student pays a common annual mess fee and is required to register with one of the two messes for a fixed period of two months. During this period the student may take meals only at the registered mess. This arrangement restricts the choice available to students and, since neither mess has advance information about the number of students who will actually attend a given meal, the quantity of food prepared is estimated rather than planned. A considerable portion of the prepared food is consequently wasted.

The present project, titled **Design and Implementation of a Hostel Mess Management System using MySQL**, addresses these limitations through a relational database application. The annual fee remains unchanged; however, for each of the four daily meals (breakfast, lunch, hi-tea and dinner) a student may select either mess, or indicate that the meal will be skipped, up to seven days in advance. Since every selection is recorded in the database, each mess is able to determine precisely how many meals are to be prepared. The quantity prepared is computed as the number of confirmed bookings together with a buffer of thirty percent for students who arrive without a booking. Attendance at each meal is recorded by the face-recognition terminals already installed at both messes, students rate the meals they have eaten, and the surplus food remaining after each meal is allocated automatically to non-governmental organisations (NGOs) operating near the campus.

The core of the system is a MySQL 8 database consisting of fifteen tables in the Third Normal Form, fifteen triggers, seven stored procedures, one stored function, eight views and one scheduled event. All significant business rules, such as the restriction of one selection per student per meal, the prohibition of bookings during a declared leave period and the requirement that a student be allotted a hostel corresponding to the student's gender, are enforced within the database itself. A web-based interface for students and an analytical dashboard for the management of both messes have been developed over the database by means of a Python (FastAPI) service; nevertheless, the logic of the application resides in SQL and can be demonstrated entirely through MySQL Workbench.

2. Objectives

The principal objectives of the project are as follows:

1. To design a normalised relational database that records students, hostels, academic programmes, messes, meals, bookings, attendance and donations without redundancy.
2. To permit every student to select a mess independently for each meal of the day, in place of a fixed two-month registration.
3. To provide each mess with an accurate forecast of the quantity of food to be prepared, derived from bookings through a database view.
4. To enforce data integrity through primary keys, foreign keys, unique keys, CHECK constraints and triggers, so that invalid data cannot be stored by any application.
5. To implement multi-step operations, such as suspending a student's meals during leave, as stored procedures executed within transactions, thereby guaranteeing atomicity.
6. To close every meal automatically at the end of its serving period by means of a scheduled database event, and to allocate the surplus food equitably among partner NGOs.
7. To collect students' ratings of the meals they have eaten, admitting a rating only for a meal actually served, so that each mess can monitor the quality of its food.
8. To furnish the management of each mess with information on revenue, food cost, profit, wastage and absenteeism.
9. To secure the database by means of parameterised queries and a database user granted only the minimum necessary privileges.

Expected outcome: a fully functional, normalised database integrated with a web application, in which every transaction performed by a user is reflected immediately in the database and every business rule is demonstrably enforced by the database management system.

3. Features and Functionality

3.1 Student Module

- **Authentication:** the student logs in with a mobile number and a six-digit one-time password. A mobile number not yet present in the database is registered automatically at the first login, which in turn invokes the enrolment trigger.
- **Menu display:** the menu of both messes is displayed for each meal of the day.
- **Meal booking:** for each meal the student selects BluDove, BluSpring or Skip, for the current day and the following seven days. A change of selection updates the existing record by means of `INSERT … ON DUPLICATE KEY UPDATE`, since a unique key permits only one record per student per meal.
- **Leave management:** the student may suspend meals for a period of leave through the stored procedure `sp_pause_mess` and may resume them early through `sp_resume_mess`.

3.2 Attendance and Automatic Closure of Meals

- **Recording of attendance:** both messes have face-recognition terminals at the serving counter. On recognising a student, the terminal inserts a record into the `attendance` table; a student served without a booking at that mess is identified as a walk-in.
- **Automatic closure of a meal:** the scheduled event `ev_close_meals` runs every ten minutes and invokes `sp_close_finished_meals`, which closes every meal whose serving period has ended by calling `sp_close_meal`. The latter computes the surplus and records its donation to an NGO. No manual action is required.

3.3 Meal Rating Module

- After a meal recorded by the face-recognition terminal, the student may rate it from one to five stars on the booking page.
- The rating is stored in the table `ratings`. A BEFORE trigger admits the rating only if a corresponding attendance record exists, and a CHECK constraint restricts the value to the range one to five.
- The view `v_mess_ratings` provides the average rating of each mess for each meal, which the management dashboard presents as a ratings board.

3.4 Management Module

- Revenue, food cost, profit, profit margin, wastage and absenteeism of each mess over the preceding seven days, obtained from the view `v_mess_earnings`.

- Bookings received for the following day and the corresponding quantity to be prepared, obtained from the view `v_daily_forecast`, for the planning of procurement.
- The average rating of each mess for each meal over the preceding seven days, obtained from the view `v_mess_ratings`.
- A meal-wise summary of the current day, the distribution of hostels between the two messes, the record of donations, and an activity log maintained exclusively by triggers.

3.5 Database Operations Employed

Category of operation	Application in the project
DDL: CREATE TABLE, VIEW, TRIGGER, PROCEDURE, FUNCTION; CREATE USER; GRANT	Definition of the schema (db/schema.sql) and of database users
DDL: ALTER TABLE, DROP; DML: INSERT, UPDATE, DELETE, INSERT … ON DUPLICATE KEY UPDATE	Login, booking, attendance, ratings, leave management
Joins, subqueries, correlated subqueries, UNION, GROUP BY, HAVING, WITH ROLLUP	Views and reports
Transactions with rollback on error and savepoints; cursors; loops	sp_pause_mess, sp_resume_mess, sp_close_meal, sp_close_finished_meals, sp_demo_week
Scheduled events (CREATE EVENT)	ev_close_meals: automatic closure of finished meals
CREATE USER, GRANT, REVOKE	Least-privilege application user; read-only management user
BEFORE and AFTER triggers; SIGNAL SQLSTATE	Enforcement of business rules and maintenance of the audit log

4. Tools and Technologies Used

Tool / Technology	Purpose
MySQL Server 8 (InnoDB storage engine)	Relational database management system used for tables, constraints, triggers, stored procedures, functions, views and users. The system has been tested on MySQL 8.0 (Windows) and 26.10.0 (macOS).
MySQL Workbench	Development and execution of SQL statements, inspection of result sets, generation of the ER diagram by reverse engineering, and demonstration of the project.
Structured Query Language (MySQL dialect)	Definition of the complete schema and business logic.
Python 3.12, FastAPI and PyMySQL	Application programming interface between the web pages and the database. All queries are parameterised and no object-relational mapper is employed, so that the SQL remains explicit.
HTML, CSS and JavaScript	User interfaces for students (booking and rating) and for the management of the messes (dashboard).
Git and GitHub	Version control and automated verification of the installation procedure on Windows.
pytest	Automated tests (twenty-three in number) verifying the rules enforced by the database.

5. ER Diagram and Relational Model

5.1 Entities and Attributes

Entity (table)	Description	Principal attributes
programmes	Academic programme of study	id (PK), code (UNIQUE), name, school, years
hostels	Hostel block; B1–B12 for male and G1–G10 for female students	id (PK), name (UNIQUE), for_gender, rooms
students	Student enrolled with the university	id (PK), reg_no, phone, email (UNIQUE), programme_id (FK), hostel_id (FK), room_no
logins	Successful login of a student	id (PK), student_id (FK), firebase_uid, at
messes	Mess (BluDove or BluSpring)	id (PK), name (UNIQUE)
meal_slots	Meal of the day with its timing	id (PK), starts_at, ends_at, paid_per_plate
mess_costs	Cost of preparing one plate at a mess for a meal	(mess_id, slot_id) composite PK; both FK
menu_items	Dish offered by a mess at a meal	id (PK), mess_id (FK), slot_id (FK), dish
bookings	Selection made by a student for a meal on a date	id (PK), UNIQUE (student_id, meal_date, slot_id), status
pauses	Period of leave of a student	id (PK), student_id (FK), from_date, to_date, ended_on
attendance	Meal actually served to a student	id (PK), UNIQUE (student_id, meal_date, slot_id), mess_id (FK)
ratings	Rating of a meal by a student who ate it	id (PK), UNIQUE (student_id, meal_date, slot_id), mess_id (FK), stars (CHECK 1–5)
ngos	Partner non-governmental organisation	id (PK), name (UNIQUE), area
donations	Surplus of a mess for a meal, donated to an NGO	id (PK), UNIQUE (meal_date, slot_id, mess_id), ngo_id (FK)
activity_log	Audit record, written exclusively by triggers	id (PK), at, event, student_id, detail

5.2 Relationships

All relationships in the model are of the one-to-many type. A programme and a hostel are each associated with many students. A student is associated with many bookings, periods of leave, attendance records, ratings and logins. A mess and a meal slot each participate in many menu items, bookings, attendance records, ratings and donations, and an NGO receives many donations. The relationship between messes and meal slots is many-to-many with respect to cost; it is resolved by the associative entity `mess_costs`, whose primary key is the composite of the two foreign keys. Each connector in the following diagram corresponds to a FOREIGN KEY constraint defined in the database.

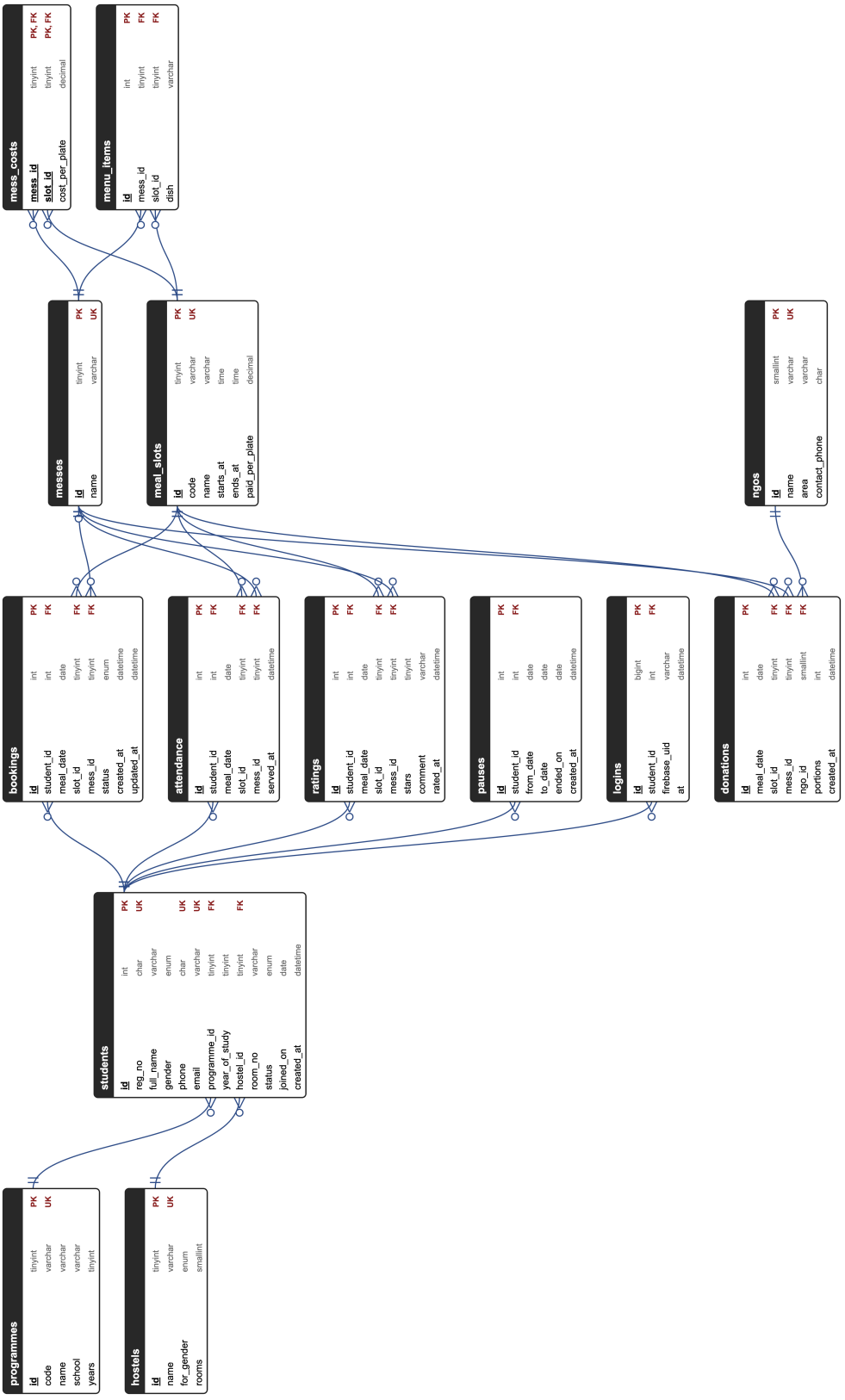


Figure 1: Entity-Relationship diagram of the database (PK: primary key, FK: foreign key, UK: unique key). The table activity_log is omitted, as it is an audit table without foreign keys.

5.3 Relational Schema

In the following schema, primary keys are underlined and foreign keys are shown in italics and marked with an asterisk (*).

activity_log(id, at, event, student_id, detail)

attendance(id, *student_id**, meal_date, *slot_id**, *mess_id**, served_at)

bookings(id, *student_id**, meal_date, *slot_id**, *mess_id**, status, created_at, updated_at)

donations(id, meal_date, *slot_id**, *mess_id**, *ngo_id**, portions, created_at)

hostels(id, name, for_gender, rooms)

logins(id, *student_id**, firebase_uid, at)

meal_slots(id, code, name, starts_at, ends_at, paid_per_plate)

menu_items(id, *mess_id**, *slot_id**, dish)

mess_costs(*mess_id**, *slot_id**, cost_per_plate)

messes(id, name)

ngos(id, name, area, contact_phone)

pauses(id, *student_id**, from_date, to_date, ended_on, created_at)

programmes(id, code, name, school, years)

ratings(id, *student_id**, meal_date, *slot_id**, *mess_id**, stars, comment, rated_at)

students(id, reg_no, full_name, gender, phone, email, *programme_id**, year_of_study, *hostel_id**, room_no, status, joined_on, created_at)

5.4 Foreign Keys

Foreign key	References
attendance.student_id	students.id
attendance.slot_id	meal_slots.id
attendance.mess_id	messes.id
bookings.student_id	students.id
bookings.slot_id	meal_slots.id
bookings.mess_id	messes.id
donations.slot_id	meal_slots.id
donations.mess_id	messes.id
donations.ngo_id	ngos.id
logins.student_id	students.id
menu_items.mess_id	messes.id

menu_items.slot_id	meal_slots.id
mess_costs.mess_id	messes.id
mess_costs.slot_id	meal_slots.id
pauses.student_id	students.id
ratings.student_id	students.id
ratings.slot_id	meal_slots.id
ratings.mess_id	messes.id
students.programme_id	programmes.id
students.hostel_id	hostels.id

5.5 Normalisation

First Normal Form (1NF): every attribute holds an atomic value. The dishes of a meal are stored as separate rows of `menu_items` rather than as a list, and the four meals of a student are stored as four rows of `bookings` rather than as four columns. **Second Normal Form (2NF):** no non-key attribute depends on a part of a composite key. In `mess_costs`, whose key is (mess_id, slot_id), the attribute cost_per_plate depends on the complete key. **Third Normal Form (3NF):** no non-key attribute depends transitively on the key. The gender of a hostel is stored only in `hostels` and is not repeated in `students`. Derived quantities, such as the quantity to be prepared, the profit and the last login of a student, are computed through views and are not stored, which eliminates the possibility of inconsistency.

6. Database Implementation

6.1 Table Creation, Data Types and Constraints

All tables employ the InnoDB storage engine, which supports foreign keys and transactions, and the `utf8mb4` character set. Data types have been selected according to the nature of the data: `TINYINT UNSIGNED` for small reference tables, `INT UNSIGNED` for students and bookings, `BIGINT` for the audit log, `CHAR(10)` for mobile numbers, `DECIMAL(6,2)` for monetary values, `DATE`, `TIME` and `DATETIME(3)` for temporal values, and `ENUM` for attributes with a fixed set of values. The definitions of the two principal tables are reproduced below.

```
CREATE TABLE students (
  id             INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  reg_no         CHAR(9) NOT NULL UNIQUE,
  full_name      VARCHAR(80) NOT NULL,
  gender         ENUM('M','F') NOT NULL,
  phone          CHAR(10) NOT NULL UNIQUE,
  email          VARCHAR(120) NOT NULL UNIQUE,
  programme_id   TINYINT UNSIGNED NOT NULL,
  year_of_study  TINYINT UNSIGNED NOT NULL,
  hostel_id      TINYINT UNSIGNED NOT NULL,
  room_no        VARCHAR(6) NOT NULL,
  status         ENUM('active','graduated','left') NOT NULL DEFAULT 'active',
  joined_on      DATE NOT NULL,
  created_at     DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  INDEX idx_student_hostel (hostel_id),
  INDEX idx_student_programme (programme_id, year_of_study),
  FOREIGN KEY (programme_id) REFERENCES programmes(id),
  FOREIGN KEY (hostel_id)    REFERENCES hostels(id),
  CONSTRAINT chk_phone CHECK (REGEXP_LIKE(phone, '^[6-9][0-9]{9}$')),
  CONSTRAINT chk_reg_no CHECK (REGEXP_LIKE(reg_no, '^[0-9]{9}$')),
  CONSTRAINT chk_room CHECK (REGEXP_LIKE(room_no, '^[0-9]{3,4}$'))
);
```

```
CREATE TABLE bookings (
  id          INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  student_id  INT UNSIGNED NOT NULL,
  meal_date   DATE NOT NULL,
  slot_id     TINYINT UNSIGNED NOT NULL,
  mess_id     TINYINT UNSIGNED NULL,
  status      ENUM('booked','skipped','paused') NOT NULL,
  created_at  DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  updated_at  DATETIME(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3) ON UPDATE
CURRENT_TIMESTAMP(3),
  UNIQUE KEY uq_one_choice (student_id, meal_date, slot_id),
  INDEX idx_forecast (meal_date, slot_id, mess_id, status),
  FOREIGN KEY (student_id) REFERENCES students(id),
  FOREIGN KEY (slot_id)    REFERENCES meal_slots(id),
  FOREIGN KEY (mess_id)    REFERENCES messes(id),
  CONSTRAINT chk_status_mess CHECK ((status = 'booked') = (mess_id IS NOT NULL))
);
```

Constraint	Instance in the database	Purpose
PRIMARY KEY	Every table; composite key in mess_costs	Unique identification of each row
FOREIGN KEY	bookings.student_id → students.id (twenty in total)	Referential integrity
UNIQUE	bookings (student_id, meal_date, slot_id)	Only one selection per student per meal
CHECK	chk_phone, chk_status_mess, chk_pause_range, chk_slot_time, chk_stars	Validity of mobile numbers, consistency of status, valid date ranges and timings, ratings from one to five
NOT NULL / DEFAULT	students.hostel_id NOT NULL; created_at DEFAULT CURRENT_TIMESTAMP(3)	Completeness of data
Triggers	trg_student_check_ins, trg_booking_paused_ins, trg_rating_check_ins	Rules involving more than one table, which cannot be expressed as CHECK constraints

6.2 Population of the Database

The installation script loads the reference data, namely two messes, four meals, 31 dishes, 12 academic programmes, 22 hostels and three NGOs, together with a roster of 1,346 students. The stored procedure `sp_demo_week` subsequently generates the activity of one week entirely in SQL. At the time of preparation of this report, the database contains 29,062 bookings, 17,665 records of meals served, 5,486 ratings, 56 donations and 62,163 audit records.

6.3 Data Manipulation Performed by the Application

```
-- Booking or modification of a meal (one row per student per meal, maintained
by the UNIQUE key)
INSERT INTO bookings (student_id, meal_date, slot_id, mess_id, status)
VALUES (%s, %s, %s, %s, %s) AS new
ON DUPLICATE KEY UPDATE mess_id = new.mess_id, status = new.status;

-- Recording of a meal served (performed by the face-recognition terminal)
INSERT INTO attendance (student_id, meal_date, slot_id, mess_id) VALUES (%s, %s,
%s, %s);

-- Rating of a meal eaten (admitted only if the attendance record exists)
INSERT INTO ratings (student_id, meal_date, slot_id, mess_id, stars, comment)
VALUES (%s, %s, %s, %s, %s, %s) AS new
ON DUPLICATE KEY UPDATE stars = new.stars, comment = new.comment;

-- Recording of a login (invokes the LOGIN trigger)
INSERT INTO logins (student_id, firebase_uid) VALUES (%s, %s);
```

The placeholders (`%s`) are bound by the database driver and are never concatenated into the SQL text; the application is therefore not susceptible to SQL injection.

7. SQL Queries and Outputs

Each query presented in this section was executed on the project database, and the result shown is the output obtained.

7.1 Data Definition and Database Structure

```
SHOW FULL TABLES
```

	Tables_in_mess	Table_type
▸	activity_log	BASE TABLE
▸	attendance	BASE TABLE
▸	bookings	BASE TABLE
▸	donations	BASE TABLE
▸	hostels	BASE TABLE
▸	logins	BASE TABLE
▸	meal_slots	BASE TABLE
▸	menu_items	BASE TABLE
▸	mess_costs	BASE TABLE
▸	messes	BASE TABLE
▸	ngos	BASE TABLE
▸	pauses	BASE TABLE
▸	programmes	BASE TABLE
▸	ratings	BASE TABLE
▸	students	BASE TABLE
▸	v_daily_forecast	VIEW
▸	v_hostel_split	VIEW
▸	v_live_bookings	VIEW
▸	v_meal_summary	VIEW
▸	v_mess_earnings	VIEW
▸	v_mess_ratings	VIEW

23 row(s) returned, first 21 shown

Figure 2: List of tables and views in the database

```
DESCRIBE students
```

	Field	Type	Null	Key	Default	Extra
▸	id	int unsigned	NO	PRI	*NULL*	auto_increm
▸	reg_no	char(9)	NO	UNI	*NULL*	
▸	full_name	varchar(80)	NO		*NULL*	
▸	gender	enum('M','F')	NO		*NULL*	
▸	phone	char(10)	NO	UNI	*NULL*	
▸	email	varchar(120)	NO	UNI	*NULL*	
▸	programme_id	tinyint unsigned	NO	MUL	*NULL*	
▸	year_of_study	tinyint unsigned	NO		*NULL*	
▸	hostel_id	tinyint unsigned	NO	MUL	*NULL*	
▸	room_no	varchar(6)	NO		*NULL*	
▸	status	enum('active','graduated','left')	NO		active	
▸	joined_on	date	NO		*NULL*	
▸	created_at	datetime(3)	NO		CURRENT_TIMESTAMP(3)	DEFAULT_C

13 row(s) returned

Figure 3: Structure of the table students

```
ALTER TABLE students ADD COLUMN blood_group VARCHAR(3) NULL AFTER gender;
SELECT column_name, column_type, is_nullable FROM
information_schema.columns
WHERE table_schema = 'mess' AND table_name = 'students' AND column_name =
'blood_group';
ALTER TABLE students DROP COLUMN blood_group
```

✔ ALTER … 0 row(s) affected

	COLUMN_NAME	COLUMN_TYPE	IS_NULLABLE
▸	blood_group	varchar(3)	YES

1 row(s) returned

✔ ALTER … 0 row(s) affected

Figure 4: DDL: addition and removal of a column with ALTER TABLE

```
SELECT (SELECT COUNT(*) FROM students) AS students,
       (SELECT COUNT(*) FROM bookings) AS bookings,
       (SELECT COUNT(*) FROM attendance) AS meals_served,
       (SELECT COUNT(*) FROM donations) AS donations,
       (SELECT COUNT(*) FROM activity_log) AS log_rows
```

	students	bookings	meals_served	donations	log_rows
▸	1346	29062	17665	56	62163

1 row(s) returned

Figure 5: Number of rows in the principal tables

7.2 Data Manipulation: Insertion, Updation and Deletion

```
INSERT INTO students (reg_no, full_name, gender, phone, email,
programme_id,
                      year_of_study, hostel_id, room_no, joined_on)
VALUES ('250105227', 'Rohan Mehta', 'M', '9876500001',
        'rohan.250105227@muj.manipal.edu', 1, 2, 4, '412', '2025-07-21');
SELECT reg_no, full_name, programme, year_of_study, hostel, room_no
FROM v_student_directory WHERE phone = '9876500001'
```

✔ INSERT … 1 row(s) affected

	reg_no	full_name	programme	year_of_study	hostel	room_no
▸	250105227	Rohan Mehta	BTECH-CSE	2	B4	412

1 row(s) returned

Figure 6: Insertion of a student record and its retrieval through a view

```
UPDATE students SET room_no = '415' WHERE phone = '9876500001';
SELECT event, detail FROM activity_log
WHERE student_id = (SELECT id FROM students WHERE phone = '9876500001')
```

✔ UPDATE … 1 row(s) affected

	event	detail
▸	ENROL	Rohan M. · BTECH-CSE year 2 · B4 room 412

1 row(s) returned

Figure 7: Updation of a record and the corresponding audit entry generated by a trigger

```
INSERT INTO ratings (student_id, meal_date, slot_id, mess_id, stars)
VALUES (1, CURDATE(), 2, 2, 4);
SELECT r.stars, ms.name AS meal, m.name AS mess, a.served_at
FROM ratings r
JOIN attendance a  ON a.student_id = r.student_id AND a.meal_date =
r.meal_date AND a.slot_id = r.slot_id
JOIN meal_slots ms ON ms.id = r.slot_id
JOIN messes m      ON m.id = r.mess_id
WHERE r.student_id = 1 AND r.meal_date = CURDATE()
```

✔ INSERT … 1 row(s) affected

	stars	meal	mess	served_at
▸	4	Breakfast	BluSpring	2026-10-27 19:41:48
▸	4	Lunch	BluSpring	2026-10-27 19:41:48

2 row(s) returned

Figure 8: Insertion of a rating for a meal that the student has eaten

```
DELETE FROM ratings
WHERE student_id = 1 AND meal_date = CURDATE() AND slot_id = 2;
SELECT COUNT(*) AS remaining FROM ratings
WHERE student_id = 1 AND meal_date = CURDATE()
```

✔ DELETE … 1 row(s) affected

	remaining
▸	1

1 row(s) returned

Figure 9: DML: deletion of a record with DELETE

7.3 Selection and Joins

```
SELECT reg_no, full_name, year_of_study, room_no
FROM students
WHERE hostel_id = (SELECT id FROM hostels WHERE name = 'G1') AND
year_of_study = 2
ORDER BY full_name
LIMIT 6
```

	reg_no	full_name	year_of_study	room_no
▸	250100464	Aadhya Meena	2	110
▸	250201150	Aarohi Khanna	2	501
▸	250700845	Anika Patel	2	606
▸	250800970	Avni Kulkarni	2	230
▸	250600936	Avni Tiwari	2	504
▸	251100295	Divya Bhatia	2	328

6 row(s) returned

Figure 10: Selection with WHERE, ORDER BY and LIMIT clauses

```
SELECT b.meal_date, ms.name AS meal, COALESCE(m.name, '-') AS mess,
b.status
FROM bookings b
JOIN students s     ON s.id = b.student_id
JOIN meal_slots ms  ON ms.id = b.slot_id
LEFT JOIN messes m  ON m.id = b.mess_id
WHERE s.phone = '7000000001'
ORDER BY b.meal_date DESC, ms.id
LIMIT 8
```

	meal_date	meal	mess	status
▸	2026-10-28	Breakfast	-	skipped
▸	2026-10-28	Lunch	-	skipped
▸	2026-10-28	Hi-Tea	-	skipped
▸	2026-10-28	Dinner	-	skipped
▸	2026-10-27	Breakfast	-	skipped
▸	2026-10-27	Lunch	BluSpring	booked
▸	2026-10-27	Hi-Tea	BluDove	booked
▸	2026-10-27	Dinner	BluSpring	booked

8 row(s) returned

Figure 11: Inner and left outer joins across four tables: bookings of a student

7.4 Grouping, Aggregation and Subqueries

```
SELECT ms.name AS meal,
       SUM(b.mess_id = 1)          AS bludove,
       SUM(b.mess_id = 2)          AS bluspring,
       SUM(b.status = 'skipped')   AS skipped,
       ROUND(100 * SUM(b.mess_id = 1) / SUM(b.status = 'booked'), 1) AS
bludove_pct
FROM bookings b JOIN meal_slots ms ON ms.id = b.slot_id
WHERE b.meal_date = CURDATE() - INTERVAL 1 DAY
GROUP BY ms.id, ms.name
ORDER BY ms.id
```

	meal	bludove	bluspring	skipped	bludove_pct
▸	Breakfast	391	336	107	53.8
▸	Lunch	295	447	69	39.8
▸	Hi-Tea	387	205	241	65.4
▸	Dinner	364	401	78	47.6

4 row(s) returned

Figure 12: Grouping with conditional aggregation: distribution of the previous day's bookings between the messes

```
SELECT h.name AS hostel, COUNT(*) AS plates_booked
FROM bookings b
JOIN students s ON s.id = b.student_id
JOIN hostels h  ON h.id = s.hostel_id
WHERE b.status = 'booked'
GROUP BY h.id, h.name
HAVING COUNT(*) > (SELECT AVG(n) FROM (SELECT COUNT(*) AS n FROM bookings
b2
                   JOIN students s2 ON s2.id = b2.student_id
                   WHERE b2.status = 'booked' GROUP BY s2.hostel_id) x)
ORDER BY plates_booked DESC
```

	hostel	plates_booked
▸	G1	2006
▸	G2	1369
▸	B8	1360
▸	B11	1341
▸	B2	1294
▸	B5	1239
▸	B7	1207
▸	B1	1184

11 row(s) returned, first 8 shown

Figure 13: HAVING clause with a nested subquery: hostels with bookings above the average

```
SELECT p.code AS programme, s.year_of_study AS yr, COUNT(*) AS students
FROM students s JOIN programmes p ON p.id = s.programme_id
WHERE p.code IN ('BTECH-CSE', 'BBA')
GROUP BY p.code, s.year_of_study WITH ROLLUP
```

	programme	yr	students
▸	BBA	1	42
▸	BBA	2	49
▸	BBA	3	37
▸	BBA	*NULL*	128
▸	BTECH-CSE	1	98
▸	BTECH-CSE	2	233
▸	BTECH-CSE	3	95
▸	BTECH-CSE	4	114
▸	BTECH-CSE	*NULL*	540
▸	*NULL*	*NULL*	668

10 row(s) returned

Figure 14: GROUP BY with ROLLUP: subtotals by programme and the grand total

```
SELECT s.reg_no, fn_short_name(s.full_name) AS student, ms.name AS meal,
m.name AS ate_at
FROM attendance a
JOIN students s    ON s.id = a.student_id
JOIN meal_slots ms ON ms.id = a.slot_id
JOIN messes m      ON m.id = a.mess_id
WHERE NOT EXISTS (SELECT 1 FROM bookings b
                  WHERE b.student_id = a.student_id AND b.meal_date =
a.meal_date
                    AND b.slot_id = a.slot_id AND b.mess_id = a.mess_id AND
b.status = 'booked')
LIMIT 6
```

	reg_no	student	meal	ate_at
▸	260600001	Anvi D.	Breakfast	BluSpring
▸	240700012	Anika N.	Breakfast	BluSpring
▸	260700016	Arnav R.	Breakfast	BluSpring
▸	261100019	Tanvi B.	Breakfast	BluSpring
▸	250100025	Siya B.	Breakfast	BluSpring
▸	240800027	Rahul M.	Breakfast	BluSpring

6 row(s) returned

Figure 15: Correlated subquery (NOT EXISTS): students served without a booking at that mess

7.5 Functions

```
SELECT full_name,
       fn_short_name(full_name)               AS public_name,
       UPPER(LEFT(full_name, 1))              AS initial,
       CONCAT('+91 ', LEFT(phone, 5), ' ', RIGHT(phone, 5)) AS
phone_pretty,
       TIMESTAMPDIFF(MONTH, joined_on, CURDATE()) AS months_at_muj,
       DATE_FORMAT(joined_on, '%d %b %Y')     AS joined
FROM students ORDER BY id LIMIT 5
```

	full_name	public_name	initial	phone_pretty	months_at_muj	joined
▸	Anvi Das	Anvi D.	A	+91 70000 00001	2	30 Jul 2026
▸	Ansh Nair	Ansh N.	A	+91 70000 00002	2	29 Jul 2026
▸	Siya Malhotra	Siya M.	S	+91 70000 00003	26	31 Jul 2024
▸	Yash Mehta	Yash M.	Y	+91 70000 00004	27	24 Jul 2024
▸	Aditi Goyal	Aditi G.	A	+91 70000 00005	3	18 Jul 2026

5 row(s) returned

Figure 16: Built-in string and date functions and the user-defined function fn_short_name

7.6 Views

The database defines eight views: `v_daily_forecast`, `v_hostel_split`, `v_live_bookings`, `v_meal_summary`, `v_mess_earnings`, `v_mess_ratings`, `v_ngo_totals`, `v_student_directory`. All computations are encapsulated in these views, so that the web application and MySQL Workbench obtain identical and current results.

```
SELECT meal, mess, booked, cook_for
FROM v_daily_forecast
WHERE meal_date = CURDATE()
ORDER BY slot_id, mess_id
```

	meal	mess	booked	cook_for
▸	Breakfast	BluDove	612	796
▸	Breakfast	BluSpring	468	609
▸	Lunch	BluDove	444	578
▸	Lunch	BluSpring	701	912
▸	Hi-Tea	BluDove	566	736
▸	Hi-Tea	BluSpring	367	478
▸	Dinner	BluDove	544	708
▸	Dinner	BluSpring	605	787

8 row(s) returned

Figure 17: View v_daily_forecast: quantity to be prepared by each mess for the current day

```
SELECT meal, mess, booked, cooked, served, no_shows, walk_ins, donate
FROM v_meal_summary
WHERE meal_date = CURDATE() - INTERVAL 1 DAY
ORDER BY slot_id, mess_id
```

	meal	mess	booked	cooked	served	no_shows	walk_ins	donate
▸	Breakfast	BluDove	391	509	336	55	0	145
▸	Breakfast	BluSpring	336	437	266	70	0	136
▸	Lunch	BluDove	295	384	251	44	0	111
▸	Lunch	BluSpring	447	582	381	66	0	168
▸	Hi-Tea	BluDove	387	504	324	63	0	148
▸	Hi-Tea	BluSpring	205	267	173	32	0	78
▸	Dinner	BluDove	364	474	316	48	0	134
▸	Dinner	BluSpring	401	522	340	61	0	151

8 row(s) returned

Figure 18: View v_meal_summary: meals served, absentees, walk-ins and surplus for donation

```
SELECT mess,
       SUM(served)    AS plates_served,
       SUM(revenue)   AS revenue,
       SUM(food_cost) AS food_cost,
       SUM(profit)    AS profit,
       ROUND(100 * SUM(profit) / SUM(revenue), 1) AS margin_pct
FROM v_mess_earnings
WHERE meal_date BETWEEN CURDATE() - INTERVAL 6 DAY AND CURDATE() - INTERVAL
1 DAY
GROUP BY mess_id, mess
```

	mess	plates_served	revenue	food_cost	profit	margin_pct
▸	BluDove	7,275.00	504,795.00	434,650.00	70,145.00	13.9
▸	BluSpring	7,305.00	548,705.00	467,134.00	81,571.00	14.9

2 row(s) returned

Figure 19: View v_mess_earnings: revenue, cost and profit of each mess over six days

```
SELECT meal, mess,
       ROUND(SUM(avg_stars * ratings) / SUM(ratings), 2) AS avg_stars,
       SUM(ratings) AS ratings, SUM(unhappy) AS one_or_two_stars
FROM v_mess_ratings
WHERE meal_date >= CURDATE() - INTERVAL 6 DAY
GROUP BY slot_id, meal, mess_id, mess
ORDER BY slot_id, mess_id
```

	meal	mess	avg_stars	ratings	one_or_two_stars
▸	Breakfast	BluDove	4.04	859	23
▸	Breakfast	BluSpring	3.72	709	38
▸	Lunch	BluDove	3.72	673	25
▸	Lunch	BluSpring	4.18	1,032.00	34
▸	Hi-Tea	BluDove	3.47	610	72
▸	Hi-Tea	BluSpring	3.96	379	6
▸	Dinner	BluDove	4.29	561	0
▸	Dinner	BluSpring	3.51	663	54

8 row(s) returned

Figure 20: View v_mess_ratings: average rating of each mess for each meal over seven days

```
SELECT hostel, mess, SUM(booked) AS plates
FROM v_hostel_split
WHERE meal_date = CURDATE() AND hostel IN ('B1', 'B2', 'G1', 'G2')
GROUP BY hostel_id, hostel, mess
ORDER BY hostel_id, mess
```

	hostel	mess	plates
▸	B1	BluDove	99
▸	B1	BluSpring	100
▸	B2	BluDove	106
▸	B2	BluSpring	110
▸	G1	BluDove	313
▸	G1	BluSpring	298
▸	G2	BluDove	106
▸	G2	BluSpring	113

8 row(s) returned

Figure 21: View v_hostel_split: distribution of hostels between the two messes

8. Advanced Database Operations

8.1 Transactions

A transaction groups a set of statements into a single atomic unit of work. In the example below, the costs of BluDove are increased within a transaction, which is then rolled back; the database is consequently restored to its previous state. Every stored procedure that modifies data (`sp_pause_mess`, `sp_resume_mess` and `sp_close_meal`) begins with `START TRANSACTION` and declares an `EXIT HANDLER` that executes `ROLLBACK` in the event of an error.

```
START TRANSACTION;
UPDATE mess_costs SET cost_per_plate = cost_per_plate * 1.10 WHERE mess_id
= 1;
SELECT slot_id, cost_per_plate FROM mess_costs WHERE mess_id = 1;
ROLLBACK;
SELECT slot_id, cost_per_plate FROM mess_costs WHERE mess_id = 1
```

✔ START … 0 row(s) affected

✔ UPDATE … 4 row(s) affected

	slot_id	cost_per_plate
▸	1	34.10
▸	2	53.90
▸	3	19.80
▸	4	66.00

4 row(s) returned

✔ ROLLBACK … 0 row(s) affected

	slot_id	cost_per_plate
▸	1	31.00
▸	2	49.00
▸	3	18.00
▸	4	60.00

4 row(s) returned

Figure 22: Transaction control: modification followed by ROLLBACK

A savepoint marks a point within a transaction to which the transaction may be partially rolled back:

```
START TRANSACTION;
UPDATE meal_slots SET paid_per_plate = paid_per_plate + 5 WHERE id = 1;
SAVEPOINT after_breakfast;
UPDATE meal_slots SET paid_per_plate = paid_per_plate + 5 WHERE id = 2;
ROLLBACK TO SAVEPOINT after_breakfast;
SELECT id, name, paid_per_plate FROM meal_slots WHERE id IN (1, 2);
ROLLBACK
```

✔ START … 0 row(s) affected

✔ UPDATE … 1 row(s) affected

✔ SAVEPOINT … 0 row(s) affected

✔ UPDATE … 1 row(s) affected

✔ ROLLBACK … 0 row(s) affected

	id	name	paid_per_plate
▸	1	Breakfast	65.00
▸	2	Lunch	85.00

2 row(s) returned

✔ ROLLBACK … 0 row(s) affected

Figure 23: Partial rollback with SAVEPOINT: only the second update is undone

8.2 Triggers

A trigger is executed automatically when a row of its table is modified. Triggers serve two purposes in this project. BEFORE triggers enforce business rules and reject invalid data by raising `SIGNAL SQLSTATE '45000'`. AFTER triggers maintain an audit trail by recording every enrolment, login, booking, modification, leave, meal served, rating and donation in the table `activity_log`.

```
CREATE TRIGGER trg_booking_paused_ins BEFORE INSERT ON bookings FOR EACH ROW
BEGIN
  IF NEW.status = 'booked' AND EXISTS (
       SELECT 1 FROM pauses p
       WHERE p.student_id = NEW.student_id
         AND NEW.meal_date BETWEEN p.from_date AND p.to_date
         AND (p.ended_on IS NULL OR NEW.meal_date < p.ended_on)) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Your mess is paused for this
date';
  END IF;
END$$
```

```
SELECT trigger_name, action_timing AS timing, event_manipulation AS event,
event_object_table AS on_table
FROM information_schema.triggers
WHERE trigger_schema = 'mess'
ORDER BY event_object_table, trigger_name
```

	TRIGGER_NAME	timing	event	on_table
▸	trg_attendance_log	AFTER	INSERT	attendance
▸	trg_booking_log_ins	AFTER	INSERT	bookings
▸	trg_booking_log_upd	AFTER	UPDATE	bookings
▸	trg_booking_paused_ins	BEFORE	INSERT	bookings
▸	trg_booking_paused_upd	BEFORE	UPDATE	bookings
▸	trg_donation_log	AFTER	INSERT	donations
▸	trg_login_log	AFTER	INSERT	logins
▸	trg_pause_log	AFTER	INSERT	pauses
▸	trg_resume_log	AFTER	UPDATE	pauses
▸	trg_rating_check_ins	BEFORE	INSERT	ratings
▸	trg_rating_check_upd	BEFORE	UPDATE	ratings
▸	trg_rating_log	AFTER	INSERT	ratings
▸	trg_student_check_ins	BEFORE	INSERT	students
▸	trg_student_check_upd	BEFORE	UPDATE	students
▸	trg_student_enrol	AFTER	INSERT	students

15 row(s) returned

Figure 24: Triggers defined in the database

```
INSERT INTO students (reg_no, full_name, gender, phone, email,
programme_id,
                      year_of_study, hostel_id, room_no, joined_on)
VALUES ('230100998', 'Wrong Hostel', 'M', '9876501235',
        'w@muj.manipal.edu', 1, 1, 13, '101', '2023-07-24')
```

✖ Error Code: 1644. This hostel is for the other gender

Figure 25: Rejection by a BEFORE INSERT trigger: allotment of a female hostel to a male student

```
INSERT INTO students (reg_no, full_name, gender, phone, email,
programme_id,
                      year_of_study, hostel_id, room_no, joined_on)
VALUES ('230100996', 'Bad Phone', 'M', '12345',
        'p@muj.manipal.edu', 1, 1, 1, '101', '2023-07-24')
```

✖ Error Code: 3819. Check constraint 'chk_phone' is violated.

Figure 26: Rejection by the CHECK constraint chk_phone: invalid mobile number

```
INSERT INTO ratings (student_id, meal_date, slot_id, mess_id, stars)
VALUES (1, CURDATE(), 4, 1, 5)
```

✖ Error Code: 1644. You can only rate a meal you have eaten

Figure 27: Rejection by the trigger trg_rating_check_ins: rating of a meal not eaten

8.3 Stored Procedures and Function

```
CREATE PROCEDURE sp_pause_mess(IN p_student INT UNSIGNED, IN p_from DATE, IN
p_to DATE)
BEGIN
  DECLARE EXIT HANDLER FOR SQLEXCEPTION BEGIN ROLLBACK; RESIGNAL; END;
  START TRANSACTION;
    INSERT INTO pauses (student_id, from_date, to_date) VALUES (p_student,
p_from, p_to);
    -- Meals already chosen in that range become 'paused' (not counted).
    UPDATE bookings SET status = 'paused', mess_id = NULL
    WHERE student_id = p_student AND meal_date BETWEEN p_from AND p_to;
  COMMIT;
END$$
```

```
SELECT routine_name, routine_type, is_deterministic
FROM information_schema.routines WHERE routine_schema = 'mess'
ORDER BY routine_type, routine_name
```

	ROUTINE_NAME	ROUTINE_TYPE	IS_DETERMINISTIC
▸	fn_short_name	FUNCTION	YES
▸	sp_check_student	PROCEDURE	NO
▸	sp_close_finished_meals	PROCEDURE	NO
▸	sp_close_meal	PROCEDURE	NO
▸	sp_demo_day	PROCEDURE	NO
▸	sp_demo_week	PROCEDURE	NO
▸	sp_pause_mess	PROCEDURE	NO
▸	sp_resume_mess	PROCEDURE	NO

8 row(s) returned

Figure 28: Stored procedures and function defined in the database

```
SELECT meal_date, slot_id, mess_id, status FROM bookings
WHERE student_id = 28 AND meal_date = CURDATE() ORDER BY slot_id
```

	meal_date	slot_id	mess_id	status
▸	2026-10-27	1	1	booked
▸	2026-10-27	2	1	booked
▸	2026-10-27	3	1	booked
▸	2026-10-27	4	*NULL*	skipped

4 row(s) returned

Figure 29: Bookings of a student before the declaration of leave

```
CALL sp_pause_mess(28, CURDATE(), CURDATE() + INTERVAL 1 DAY);
SELECT meal_date, slot_id, mess_id, status FROM bookings
WHERE student_id = 28 AND meal_date = CURDATE() ORDER BY slot_id
```

✔ CALL … 0 row(s) affected

	meal_date	slot_id	mess_id	status
▸	2026-10-27	1	*NULL*	paused
▸	2026-10-27	2	*NULL*	paused
▸	2026-10-27	3	*NULL*	paused
▸	2026-10-27	4	*NULL*	paused

4 row(s) returned

Figure 30: Execution of the stored procedure sp_pause_mess

```
INSERT INTO bookings (student_id, meal_date, slot_id, mess_id, status)
VALUES (28, CURDATE(), 4, 1, 'booked')
ON DUPLICATE KEY UPDATE mess_id = 1, status = 'booked'
```

✖ Error Code: 1644. Your mess is paused for this date

Figure 31: Rejection of a booking during leave by a trigger (SIGNAL SQLSTATE '45000')

```
CALL sp_resume_mess(28, CURDATE());
SELECT event, detail FROM activity_log WHERE student_id = 28 ORDER BY id
DESC LIMIT 4
```

✔ CALL … 0 row(s) affected

	event	detail
▸	CHANGE	Divya M. · Dinner · 2026-10-27 · paused → skipped
▸	CHANGE	Divya M. · Hi-Tea · 2026-10-27 · paused → skipped
▸	CHANGE	Divya M. · Lunch · 2026-10-27 · paused → skipped
▸	CHANGE	Divya M. · Breakfast · 2026-10-27 · paused → skipped

4 row(s) returned

Figure 32: Execution of sp_resume_mess and the resulting audit entries

The procedure `sp_close_meal` employs a cursor to iterate over the surplus of each mess for a meal and assigns it, within a single transaction, to the NGO that has received the least quantity so far:

```
CALL sp_close_meal(CURDATE() - INTERVAL 1 DAY, 2);
SELECT n.name AS ngo, n.area, d.portions, m.name AS mess
FROM donations d JOIN ngos n ON n.id = d.ngo_id JOIN messes m ON m.id =
d.mess_id
WHERE d.meal_date = CURDATE() - INTERVAL 1 DAY AND d.slot_id = 2
```

✔ CALL … 0 row(s) affected

	ngo	area	portions	mess
▸	Sahyog Food Shelf	Bagru	111	BluDove
▸	Ann Seva Night Shelter	Sanganer	168	BluSpring

2 row(s) returned

Figure 33: Execution of the stored procedure sp_close_meal: allocation of surplus to NGOs

8.4 User Privileges

The application does not connect to the database as the root user. The installation script creates the user `mess_app` with only those privileges that the application requires; consequently, the application is unable to delete records or to alter or drop any object. In addition, a read-only user is created for the management of the messes, with access restricted to selected views; the REVOKE statement withdraws a privilege previously granted.

```
SHOW GRANTS FOR 'mess_app'@'localhost'
```

	Grants for mess_app@localhost
▸	GRANT USAGE ON *.* TO `mess_app`@`localhost`
▸	GRANT SELECT, INSERT, UPDATE, EXECUTE ON `mess`.* TO `…

2 row(s) returned

Figure 34: Privileges granted to the application user mess_app

```
DROP USER IF EXISTS 'mess_viewer'@'localhost';
CREATE USER 'mess_viewer'@'localhost' IDENTIFIED BY 'Viewer#2026';
GRANT SELECT ON mess.v_daily_forecast TO 'mess_viewer'@'localhost';
GRANT SELECT ON mess.v_mess_earnings TO 'mess_viewer'@'localhost';
GRANT SELECT ON mess.v_mess_ratings TO 'mess_viewer'@'localhost';
REVOKE SELECT ON mess.v_mess_earnings FROM 'mess_viewer'@'localhost';
SHOW GRANTS FOR 'mess_viewer'@'localhost'
```

✔ DROP … 0 row(s) affected

✔ CREATE … 0 row(s) affected

✔ GRANT … 0 row(s) affected

✔ GRANT … 0 row(s) affected

✔ GRANT … 0 row(s) affected

✔ REVOKE … 0 row(s) affected

	Grants for mess_viewer@localhost
▸	GRANT USAGE ON *.* TO `mess_viewer`@`localhost`
▸	GRANT SELECT ON `mess`.`v_daily_forecast` TO `mess_viewer`…
▸	GRANT SELECT ON `mess`.`v_mess_ratings` TO `mess_viewer`…

3 row(s) returned

Figure 35: Creation of a read-only user, GRANT of privileges on views and REVOKE of one privilege

8.5 Scheduled Events

An event is a task executed by the MySQL event scheduler at defined intervals. The event `ev_close_meals` runs every ten minutes and invokes `sp_close_finished_meals`, which uses a cursor to select every meal of the current day whose serving period has ended and that has not yet been closed, and calls `sp_close_meal` for each. The surplus food is thereby allocated to NGOs without any manual intervention.

```
CREATE EVENT ev_close_meals ON SCHEDULE EVERY 10 MINUTE
COMMENT 'Closes finished meals and hands leftovers to NGOs'
DO CALL sp_close_finished_meals()$$
```

```
SELECT event_name, interval_value, interval_field, status, event_definition
FROM information_schema.events WHERE event_schema = 'mess';
CALL sp_close_finished_meals();
SELECT ms.name AS meal, m.name AS mess, n.name AS ngo, d.portions
FROM donations d JOIN meal_slots ms ON ms.id = d.slot_id
JOIN messes m ON m.id = d.mess_id JOIN ngos n ON n.id = d.ngo_id
WHERE d.meal_date = CURDATE() ORDER BY d.slot_id, d.mess_id
```

	EVENT_NAME	INTERVAL_VALUE	INTERVAL_FIELD	STATUS	EVENT_DEFINITION
▸	ev_close_meals	10	MINUTE	ENABLED	CALL sp_close_finished_meals(

1 row(s) returned

✔ CALL … 0 row(s) affected

	meal	mess	ngo	portions
▸	Breakfast	BluDove	Ann Seva Night Shelter	87
▸	Breakfast	BluSpring	Ann Seva Night Shelter	62
▸	Lunch	BluDove	Dahmi Community Kitchen	79
▸	Lunch	BluSpring	Dahmi Community Kitchen	118
▸	Hi-Tea	BluDove	Sahyog Food Shelf	277
▸	Hi-Tea	BluSpring	Sahyog Food Shelf	170
▸	Dinner	BluDove	Ann Seva Night Shelter	442
▸	Dinner	BluSpring	Dahmi Community Kitchen	481

8 row(s) returned

Figure 36: Scheduled event ev_close_meals and the procedure it executes

9. Screenshots

The following screenshots depict the system in operation. The student interface and the management dashboard operate on the MySQL database described in the preceding sections.

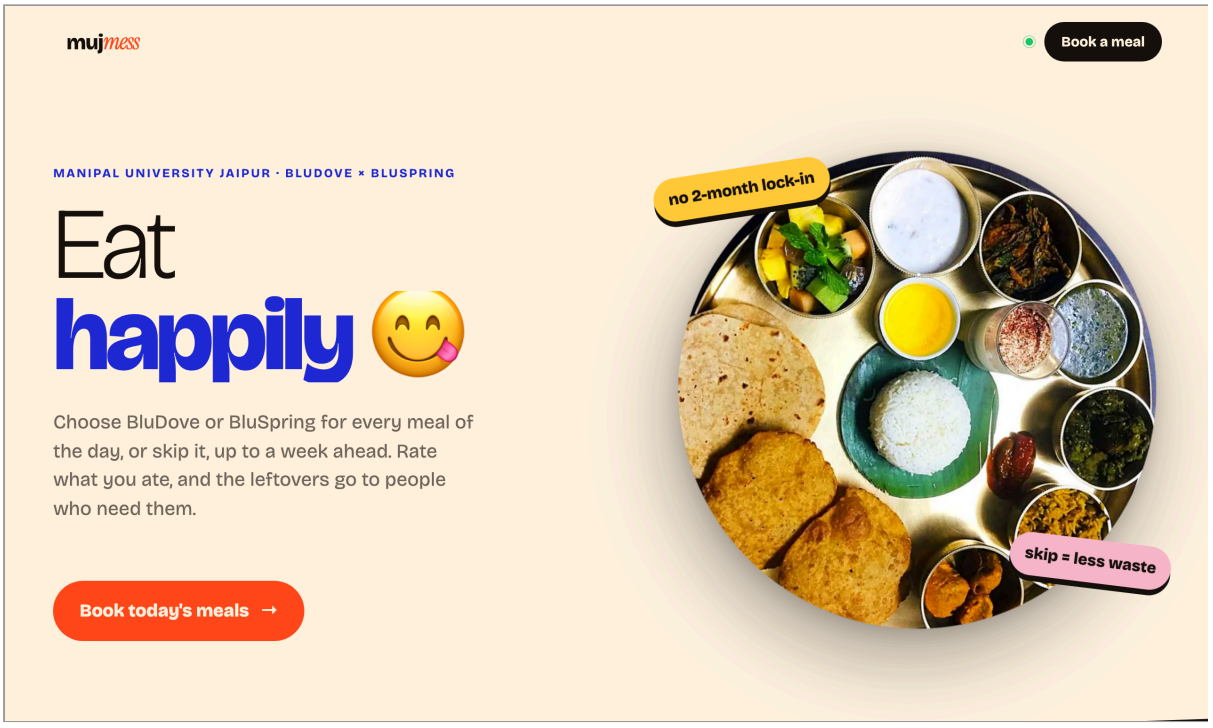


Figure 37: Home page of the student interface

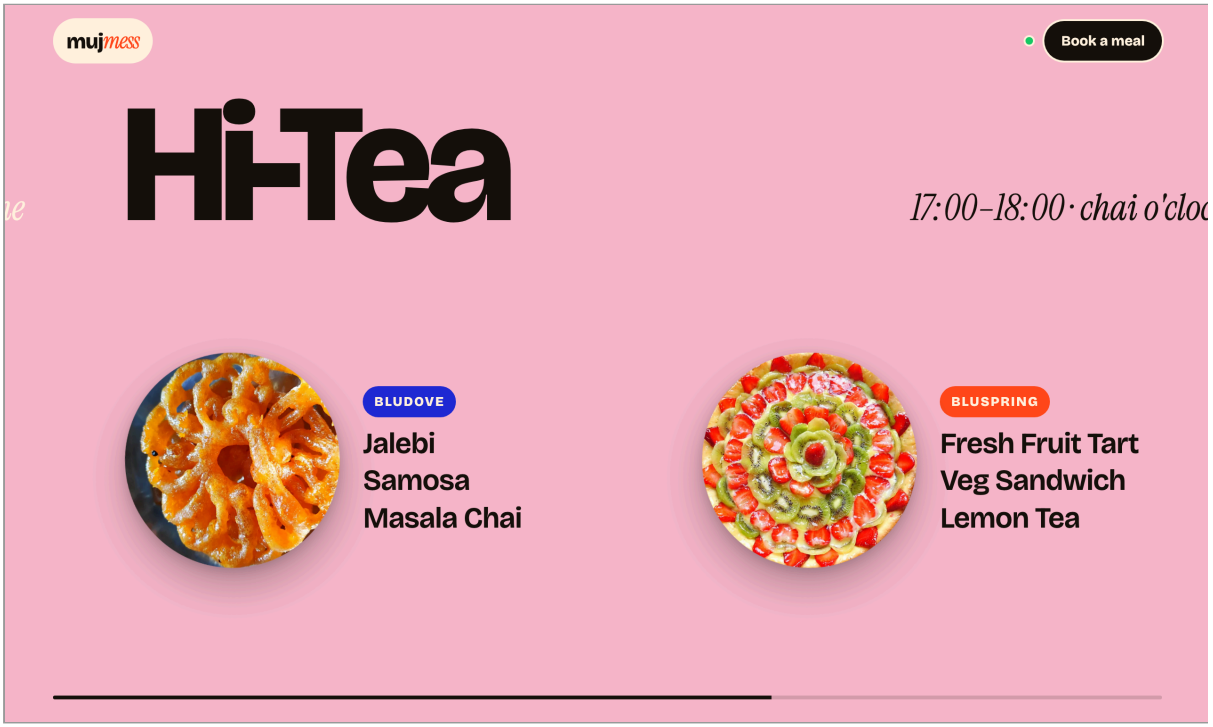


Figure 38: Menu of both messes for a meal (hi-tea)

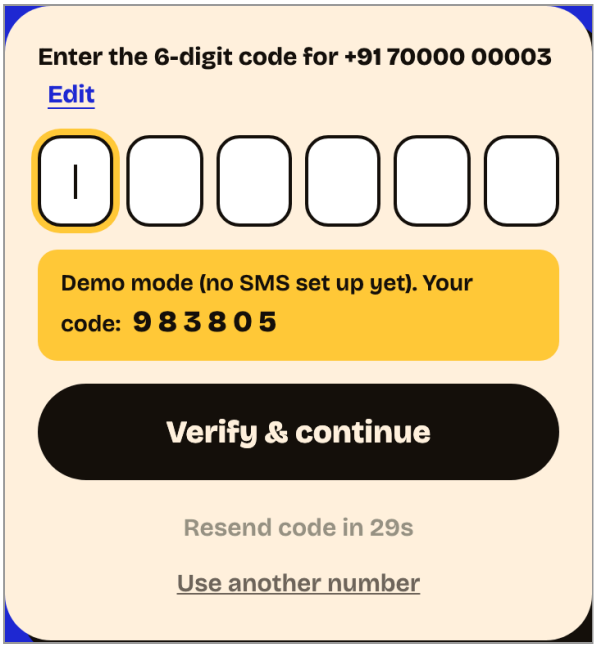


Figure 39: Login by one-time password

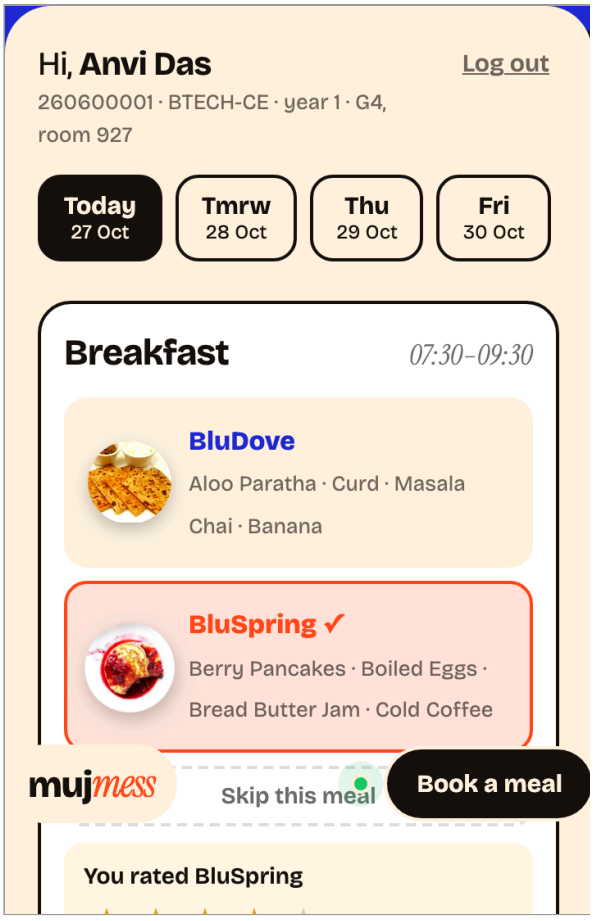


Figure 40: Meal booking screen with the student profile

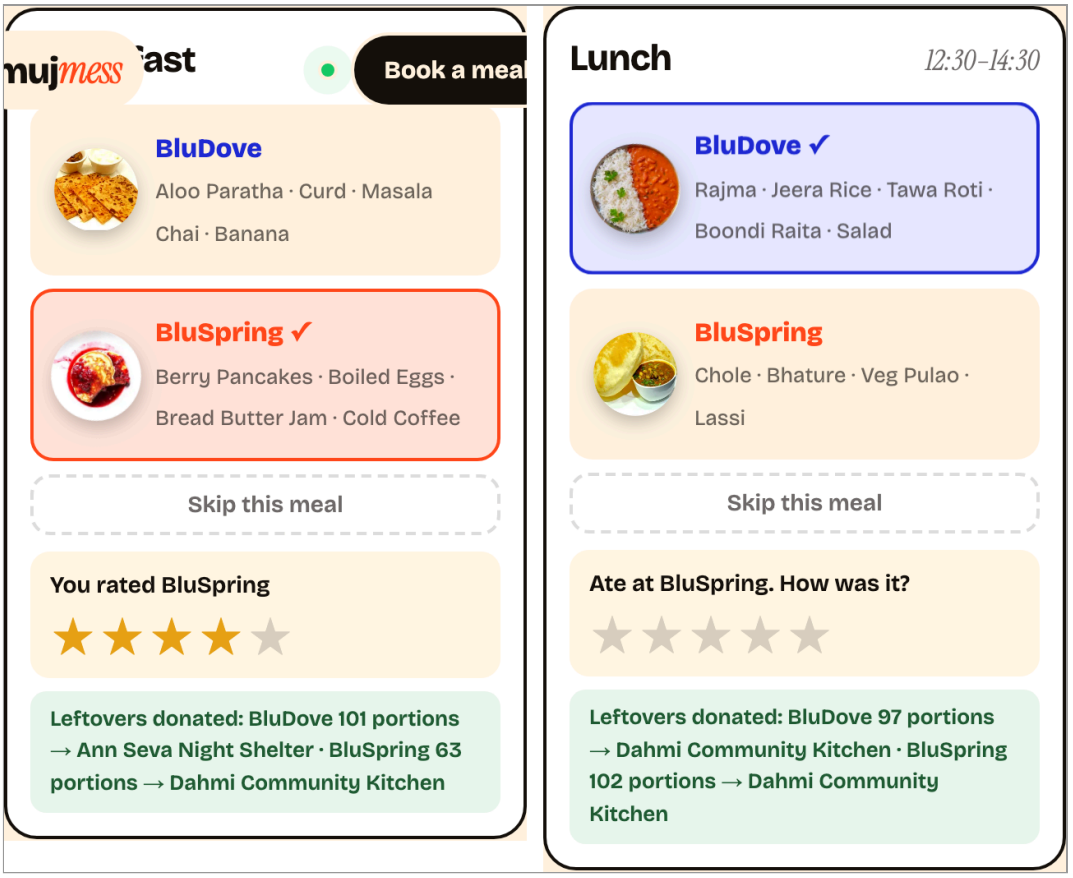
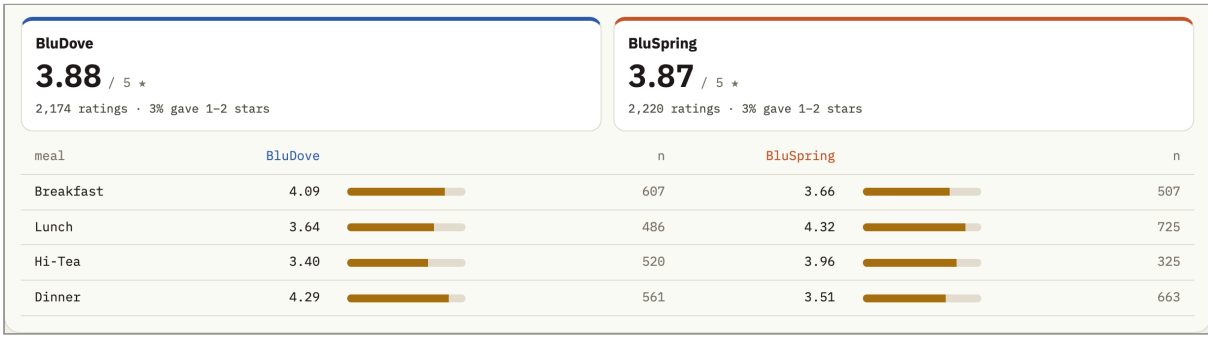


Figure 41: Rating of meals eaten (a rated breakfast, an unrated lunch) and the donation of leftovers recorded by the scheduled event



meal	BluDove	n	BluSpring	n
Breakfast	4.09	607	3.66	507
Lunch	3.64	486	4.32	725
Hi-Tea	3.40	520	3.96	325
Dinner	4.29	561	3.51	663

Figure 42: Ratings board of the management dashboard, obtained from v_mess_ratings

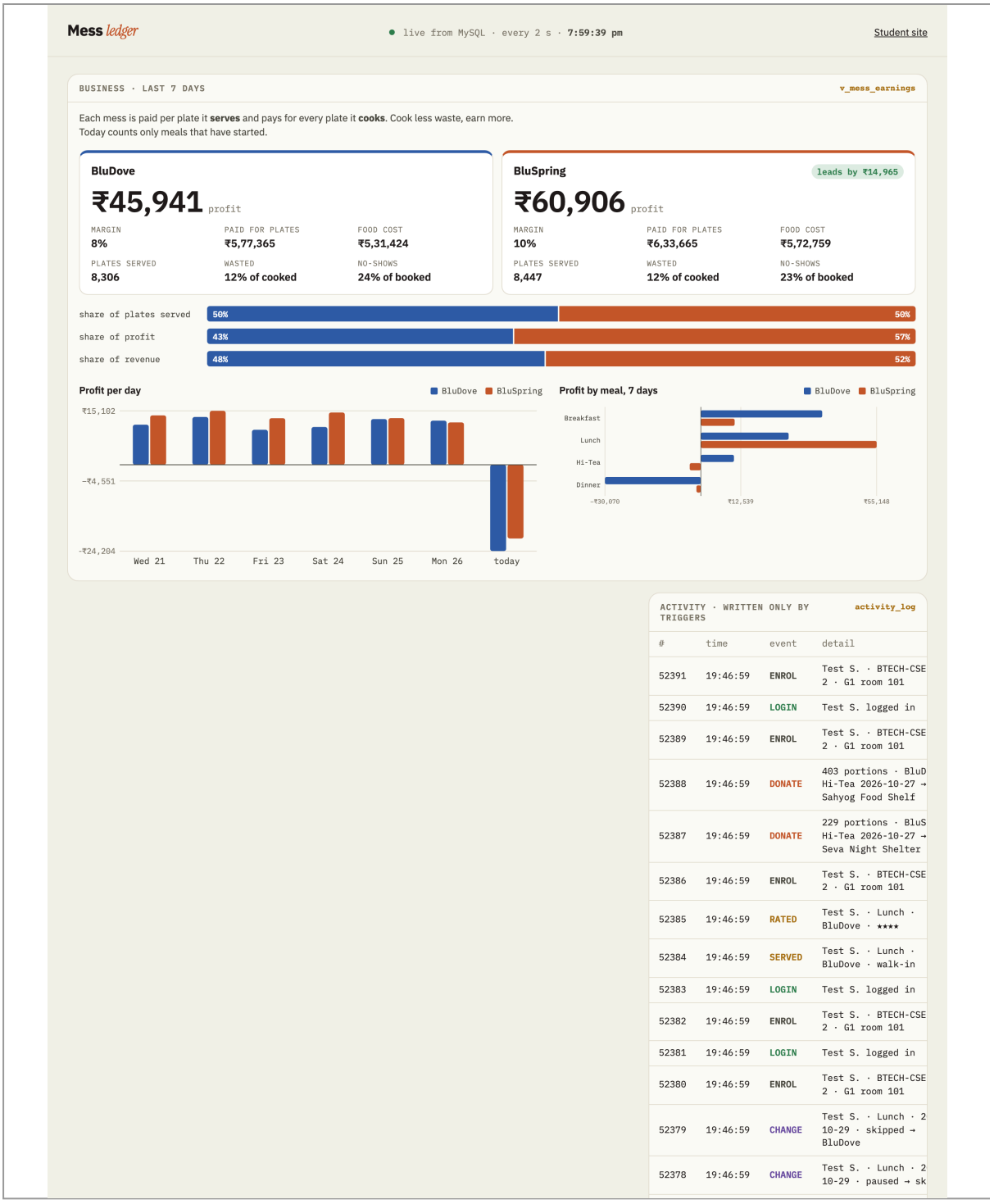


Figure 43: Management dashboard: profit of each mess (v_mess_earnings), forecast for the following day (v_daily_forecast) and the activity log

```
SELECT id, DATE_FORMAT(at, '%H:%i:%s') AS time, event, detail
FROM activity_log ORDER BY id DESC LIMIT 8
```

	id	time	event	detail
▸	62177	22:03:18	DONATE	111 portions · BluDove Lunch 2026-10-26 → Sahyog Food Shelf
▸	62176	22:03:18	DONATE	168 portions · BluSpring Lunch 2026-10-26 → Ann Seva Night Sh…
▸	62175	22:03:18	CHANGE	Divya M. · Dinner · 2026-10-27 · paused → skipped
▸	62174	22:03:18	CHANGE	Divya M. · Hi-Tea · 2026-10-27 · paused → skipped
▸	62173	22:03:18	CHANGE	Divya M. · Lunch · 2026-10-27 · paused → skipped
▸	62172	22:03:18	CHANGE	Divya M. · Breakfast · 2026-10-27 · paused → skipped
▸	62171	22:03:18	RESUME	Divya M. back from 2026-10-27
▸	62170	22:03:18	PAUSED	Divya M. · Dinner · 2026-10-27 · skipped → paused

8 row(s) returned

Figure 44: Audit log maintained by triggers

10. Conclusion

The project has replaced a fixed two-month mess registration with a per-meal choice and has substituted estimation in the kitchens with an exact forecast derived from bookings. The implementation demonstrates that a well-designed relational database can support the entire application: the web interface and the application programming interface constitute thin layers, while the rules, the computations and the history of all operations are maintained within MySQL.

Several lessons were learnt in the course of the implementation. Normalisation to the Third Normal Form eliminated the possibility of inconsistent data, and the use of views for derived quantities ensured correct results without additional program code. Constraints and triggers proved to be the most reliable means of enforcing business rules, since no client application can bypass them. Stored procedures executed within transactions ensured that multi-step operations remain consistent even in the event of failure. The principle of least privilege and the use of parameterised queries made the system secure by design. The project further afforded practical experience with features such as `ON DUPLICATE KEY UPDATE`, correlated subqueries, `WITH ROLLUP` and cursors.

The system may be extended in future by introducing a date-dependent weekly menu, a direct interface for the face-recognition terminals, and menu planning informed by the ratings collected from students.

11. References

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